

# Git Basics Workshop - Session Summary

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Repository: [github.com/kamalsingh-engg/git\\_basic\\_course](https://github.com/kamalsingh-engg/git_basic_course)

Project: Flask Team Directory (dummy employee data)

Date: September 4, 2026 | Duration: ~3 hours (Blocks 1-9)

## 1. PROJECT OVERVIEW

We built a Flask web app showing 5 dummy employees. It was used to practice Git and GitHub end-to-end.

Key files: app.py, data.py, templates/, static/style.css

.github/workflows/ci.yml - GitHub Actions CI

WORKSHOP.md + artifacts/ - guides and cheat sheets

## 2. BLOCKS COMPLETED

- Block 1 - Setup & first commit (git init, add, commit)
- Block 2 - GitHub remote, push, pull
- Block 3 - Branching (feature/add-role-badge)
- Block 4 - Pull Request #1 merged (role badges)
- Block 5 - Pull Request #2 merged (footer)
- Block 6 - Merge conflict in data.py (COMPANY\_NAME)
- Block 7 - checkout, switch, detached HEAD, restore
- Block 8 - reset, revert, go back to previous commit
- Block 9 - GitHub Actions CI + Pull Request #3 merged

## 3. FINAL COMMIT HISTORY

```
8fa5095 - Merge PR #3 (GitHub Actions + extended docs)
3f14990 - Resolve company name conflict
5d33a90 - Merge PR #2 (footer)
9336b9c - Merge PR #1 (role badges)
f1573ac - Initial commit
```

## 4. GIT COMMANDS LEARNED

- git init, status, add, commit, log
- git remote, push, pull
- git switch, branch, merge
- git restore, restore --source=HASH
- git switch --detach HASH (detached HEAD)
- git reset --soft HEAD~1 (local undo)
- git revert HASH (safe undo on shared branches)
- git stash / git stash pop

## 5. GITHUB CONCEPTS

- Remote (origin) - GitHub copy of repo

- Pull Request - review before merging to main
- Branch protection - require PR for main
- GitHub Actions - auto-test on every PR
- Merge conflict - same line edited on two branches

## 6. STANDARD WORKFLOW

feature branch -> add -> commit -> push -> open PR

CI runs (green check) -> merge PR -> git pull on main

## 7. WHEN TO USE WHAT

- Wrong file only -> git restore --source=HASH file
- Undo local commit -> git reset --soft HEAD~1
- Undo pushed commit -> git revert HASH (via PR)
- Team merge to main -> always Pull Request
- Auto-test on PR -> GitHub Actions (.github/workflows/ci.yml)

## 8. SUCCESS CRITERIA - ALL MET

- Flask app runs | 3 PRs merged | conflict resolved
- checkout/restore practiced | reset/revert understood
- GitHub Actions CI enabled and passing on main

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*Generated from Cursor AI-guided Git Basics Workshop.*